

RÉPUBLIQUE ALGÉRIENNE DÉMOCRATIQUE ET POPULAIRE  
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## CONTENTS

# CHAPTER 1

## INTRODUCTION

### 1.1 Purpose

This document provides a comprehensive overview of the platform designed to facilitate interaction and management within an educational institution. It outlines the system's objectives, functionalities, and requirements to guide development and implementation.

### 1.2 Product Scope

The platform will connect students, teachers, and administration. It provides students access to their grades, course information, and updates. Additionally, it incorporates AI-driven homework management, offering time completion estimation to help students manage their workload efficiently.

### 1.3 Glossary

- **SRS (Software Requirements Specification):** A document detailing the functional and non-functional requirements of a software system.
- **DB (Database):** A collection of electronically organized data used by the system to manage user information, courses, and grades.
- **CRUD (Create, Read, Update, Delete):** The basic operations used to manage data in a database.

- **AI (Artificial Intelligence):** The simulation of human intelligence processes by machines, particularly in the context of homework assistance.
- **User Management Module:** The component of the system that handles authentication and profiles for students and teachers.
- **Course Management:** A feature allowing teachers to manage courses while students can view their courses .

### 1.4 References

-Educational institution policies and grading rules.

-Technical references for technologies and frameworks used (e.g., databases, AI integration).

### 1.5 Overview

This SRS will detail the functionalities, system architecture, external interfaces, and both functional and non-functional requirements of the platform.

## CHAPTER 2

## GENERAL DESCRIPTION

### 2.1 Product Perspective

The platform will be a web-based application integrating several services to create an efficient academic management system. The key components include:

#### 2.1.1 User Management Module

Handles student, teacher, and administration profiles.

#### 2.1.2 Course Management

Teachers input grades, students view course information.

#### 2.1.3 Homework Management

AI-driven homework management will assist students by estimating time needed for completion based on task complexity and provide useful tips for improvement.

### 2.2 Product Function

#### 2.2.1 Course Management

Teachers can create, update, and manage courses and course-related materials.

### **2.2.2 Homework Management**

Students can submit homework through the platform, receive AI-generated time estimations for completion, and access recommendations for better performance.

## **2.3 User Characteristics**

### **2.3.1 Students**

Users who view courses, and utilize the AI-driven homework management tool to plan their study schedules.

### **2.3.2 Teachers**

Users who manage course details and receive analytics on homework submissions.

## **2.4 Constraints**

- Data privacy and security regulations (e.g., GDPR compliance).
- System must support high availability and load balancing to handle multiple users simultaneously.

## **2.5 Assumptions and Dependencies**

- The institution provides the necessary grading rules and policies.
- Availability of a stable internet connection for accessing the platform.

## CHAPTER 3

## SPECIFIC REQUIREMENTS

### 3.1 Functional Requirements

#### 3.1.1 Course Management

- Teachers can create, update, and delete courses.
- Students can view their enrolled courses.

#### 3.1.2 Homework Management

- The platform uses AI to analyze homework submissions and estimate completion times.
- Students can receive feedback on their progress.

### 3.2 Non-Functional Requirements

#### 3.2.1 Performance Requirements

- The platform must handle at least 1,000 concurrent users.

#### 3.2.2 Security Requirements

- Data encryption for all sensitive information (e.g., grades, user credentials).
- Regular security audits to ensure data protection and system integrity.

### **3.2.3 Usability Requirements**

- The platform should have a user-friendly interface accessible on smartphones.
- Accessibility compliance to accommodate users with disabilities.

### **3.2.4 Reliability Requirements**

The platform should maintain 99.9% uptime.

A backup system should ensure data recovery in case of failure.



## CHAPTER 4

## EXTERNAL INTERFACE REQUIREMENTS

### 4.1 User Interfaces

A dashboard for students, teachers, and administrators with role-specific features.

### 4.2 Hardware Interfaces

The platform will be accessible via standard web browsers on any compatible device.

### 4.3 Software Interfaces

-Integration with databases for data storage (MongoDB).

## CHAPTER 5

## FUTURE WORKS

### 5.1 AI Chatbot

Implement an AI chatbot to answer student and teacher queries in real-time, providing quick solutions to common problems.

### 5.2 Authentication with Digital ID Cards

Integrate a secure authentication system using digital ID cards to enhance user identification and data security.

## CHAPTER 6

## TOOLS

### 6.1 Frontend

We used Flutter for making this application for:

- Cross-Platform Development
- Fast Development
- Open Source

### 6.2 Backend

We used Node.js, MongoDB for our database, JWT, and testing endpoints using Postman.